

■ Zynema – AI-Powered Movie Recommendation System

Introduction

Zynema is an AI-powered movie recommendation system developed using Machine Learning and Streamlit. The project recommends movies similar to a user's selected movie based on similarity scores generated using collaborative filtering techniques. It also displays movie posters, ratings, and overviews using the TMDB API.

Abstract

The main objective of this project is to help users discover movies based on their interests. A recommendation engine was built using movie ratings datasets and cosine similarity. The application was deployed using Streamlit Cloud, providing an interactive and user-friendly interface accessible online.

Tools Used

- Python
 - Pandas
 - Scikit-Learn
 - Streamlit
 - Joblib
 - TMDB API
 - Git & GitHub
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Steps Involved in Building the Project

1. Collected movie and ratings datasets.
 2. Preprocessed and merged datasets using Pandas.
 3. Created a user-movie matrix using pivot tables.
 4. Applied cosine similarity to calculate movie similarity scores.
 5. Saved the trained similarity model using Joblib.
 6. Built an interactive UI using Streamlit.
 7. Integrated TMDB API for posters, ratings, and movie details.
 8. Deployed the application using GitHub and Streamlit Cloud.
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Conclusion

Zynema successfully demonstrates the implementation of a movie recommendation system using Machine Learning techniques. The project combines data processing, recommendation algorithms, API integration, and web deployment to provide a complete real-world AI application with an attractive user interface.
